七. $A=(a_{ij})_{n\times n}$, A^*

(1) 当 $r_n=n$ 时, 则 $r_{A^*}=n$

(2)

$\because AA^*=|A|E_n=0$

又 $\because r_n=n-1$, $\therefore A^*\neq 0$

$\therefore r_{A^*}\geq 1$

$\because r_n=n-1$, $\therefore AX=0$ 的基础解析 仅包含 一个向量

A^* ... 的每一列都是 $AX=0$ 的解

$\therefore r_{A^*}\leq 1$

$$\begin{cases} x_1 + 2x_2 - x_3 = 3 \\ x_4 = -1 \\ x_5 = 2 \end{cases}$$ 特解为 $\begin{pmatrix} 3 \\ 0 \\ 0 \\ -1 \\ 2 \end{pmatrix}$ 对应的齐次方程的基础解析为

$\alpha_1 = \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ $\alpha_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$

∴通解为 $\begin{pmatrix} 3 \\ 0 \\ 0 \\ -1 \\ 2 \end{pmatrix} + k_1 \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$

六. 设 $r_n = r$ $A = (a_{ij})_{m\times n}$，A 可以是 r 个秩为1的矩阵和

∴ $A = P \begin{pmatrix} E_r & 0 \\ 0 & 0 \end{pmatrix} Q$，其中P为m阶可逆阵，Q为n阶矩阵

$= P(E_{11} + E_{22} + E_{rr})Q$，其中 E_{ii} 为 $m\times n$ 阶矩阵，(i,i) 为1，其余位置为0

(2) $A = (a_{ij})_{m\times n}$，$r_n = n$

∴ $A = P \begin{pmatrix} E_n \\ 0 \end{pmatrix} Q$ 且 $B = Q^{-1}(E_n, 0)P^{-1}$

法一：$\begin{pmatrix} A & C \\ 0 & B \end{pmatrix} \xrightarrow{C_2 - CA^{-1}C_1} \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$

法二：eP $\begin{pmatrix} A & C \\ 0 & B \end{pmatrix}\begin{pmatrix} E_n & -A^{-1} \\ 0 & E_n \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$

法三：$\begin{pmatrix} A & C \\ 0 & B \end{pmatrix}$ 初等行变换 $\begin{pmatrix} E_n & C' \\ 0 & B \end{pmatrix} \longrightarrow \begin{pmatrix} E_n & 0 \\ 0 & B \end{pmatrix}$

四. $\begin{pmatrix} 0 & 0 & \cdots & 0 & -a_n \\ 1 & 0 & \cdots & 0 & -a_{n-1} \\ 0 & 1 & \cdots & 0 & -a_{n-2} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & 1 & -a_1 \end{pmatrix}$ 求逆矩阵

五. $\begin{cases} x_1 + 2x_2 - x_3 + 3x_4 + x_5 = 2 \\ 2x_1 + 4x_2 - 2x_3 + 6x_4 + 3x_5 = 6 \\ -x_1 - 2x_2 + x_3 - x_4 + 3x_5 = 4 \end{cases}$

$\bar{A} = \begin{bmatrix} 1 & 2 & -1 & 3 & 1 & 2 \\ 2 & 4 & -2 & 6 & 3 & 6 \\ -1 & -2 & 1 & -1 & 3 & 4 \end{bmatrix}$

$\xrightarrow[R_3 + R_1]{R_2 - 2R_1} \begin{bmatrix} 1 & 2 & -1 & 3 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 2 & 4 & 6 \end{bmatrix} \xrightarrow[R_{23}]{\frac{1}{2}R_3} \begin{bmatrix} 1 & 2 & -1 & 3 & 1 & 2 \\ 0 & 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$

$\xrightarrow[R_1 - 3R_1]{R_2 - 2R_3} \begin{bmatrix} 1 & 2 & -1 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_{13} - R_3} \begin{bmatrix} 1 & 2 & -1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$

原方程等价于

二. $(\alpha_1^T, \alpha_2^T, \alpha_3^T, \alpha_4^T, \alpha_5^T) =$

求极大线性无关组，并表示剩下的

$$\begin{pmatrix} -1 & 2 & 0 & 1 & 4 \\ 1 & 1 & 1 & 3 & 3 \\ -1 & 0 & 2 & 1 & -6 \\ -2 & -1 & 2 & -1 & -11 \\ -1 & 3 & -1 & 1 & 9 \end{pmatrix} \xrightarrow[R_5+R_1]{\substack{R_2+R_1 \\ R_3-R_1 \\ R_4-2R_1}} \cdots$$

$$\longrightarrow \begin{pmatrix} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

极大线性无关组为 $\alpha_1, \alpha_2, \alpha_3$

且 $\alpha_4 = \alpha_1 + \alpha_2 + \alpha_3$

$\alpha_5 = 2\alpha_1 + 3\alpha_2 - 2\alpha_3$

三. 证 $r\begin{pmatrix} A & B \\ 0 & B \end{pmatrix} = r(A) + r(B)$

法一：$\begin{pmatrix} A & B \\ 0 & B \end{pmatrix} \xrightarrow{R_1 - R_2} \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$ $\therefore r\begin{pmatrix} A & B \\ 0 & B \end{pmatrix} = r\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} = r_A + r_B$

法二：$\begin{pmatrix} E_n & -E_n \\ 0 & E_n \end{pmatrix}\begin{pmatrix} A & B \\ 0 & B \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$

(2) $A_{m\times n}$ 可逆矩阵

$$r\begin{pmatrix} A & C \\ 0 & B \end{pmatrix} = r(A) + r(B)$$

一.

(1) $D_n = \begin{vmatrix} a+b & ab & & & \\ 1 & a+b & ab & & \\ 0 & \ddots & \ddots & \ddots & \\ & \ddots & \ddots & a+b & ab \\ & & & 1 & a+b \end{vmatrix} = (a+b)D_{n-1} - abD_{n-2}$

$\begin{cases} D_n = (a+b)D_{n-1} - abD_{n-2} \\ D_1 = a+b \\ D_2 = (a+b)^2 - ab = a^2 + ab + b^2 \end{cases}$

$D_n - aD_{n-1} = b(D_{n-1} - aD_{n-2}) = b^{n-2}(D_2 - aD_1) = b^n$

$D_n - bD_{n-1} = a(D_{n-1} - bD_{n-2}) = a^{n-2}(D_2 - bD_1)a^n$

法二: $x^2 - (a+b)x + ab = 0 \quad x_1 = a, \ x_2 = b \quad D_n = C_1 a^n + C_2 b^n$

$D_n = C_1 a^n + C_2 n a^n$

$\therefore D_n = \frac{a^{n+1} - b^{n+1}}{a-b}$

(2) $B_n = \begin{vmatrix} a_1 & a_2 & a_3 & \cdots & a_{n-1} & a_n \\ -1 & x & 0 & \cdots & 0 & 0 \\ 0 & -1 & x & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & x & 0 \\ 0 & 0 & 0 & \cdots & -1 & x \end{vmatrix}$

$\begin{matrix} C_{n-1} + \frac{1}{x} C_n \\ C_{n-2} + \frac{1}{x} C_{n-1} \\ \vdots \\ C_1 + \frac{1}{x} C_2 \end{matrix}$ $a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_n$

当 $x = 0$ 时, $B_n = a_n (-1)^{n+1} (-1)^{n-1} = a_n$